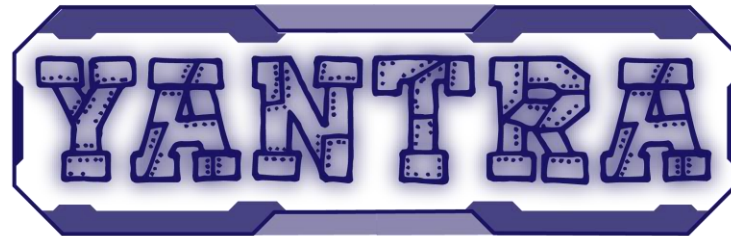


Direct Rapid Tooling



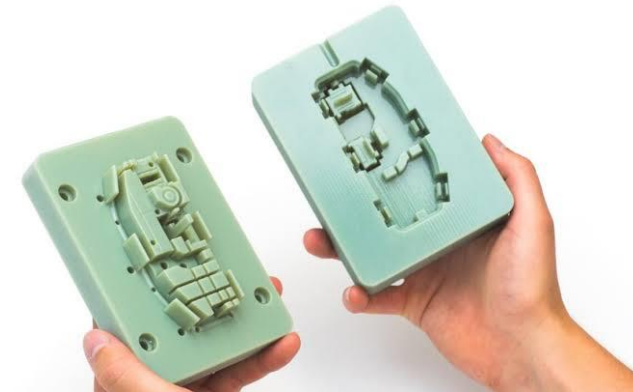
Direct AIM (ACES Injection Moulding)

- Direct AIM (ACES Injection Moulding) is a new and vitally important rapid tooling process to quickly and inexpensively build prototype parts using a variety of engineering **thermoplastics** in a **very short time** (as short as one week) without the need for production tooling.
- The process involves growing the **mold** on a **SLA system**, using the ACES build style, that is shelled out on the bottom side.
- This leaves a cavity in the mold halves that can be backfilled with various materials.
- These material include **aluminum-filled epoxy, ceramics, and low-melting metals.**

Thermoplastics



Heated & Re-Moulded 100's of times!
... can also be recycled!

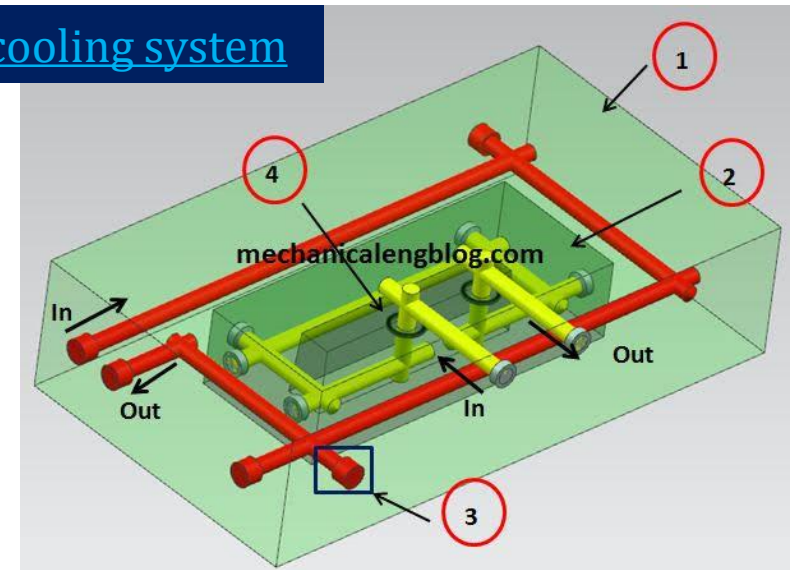


SLA mold

ACES
Accurate clear epoxy solid

- The backfilling process provides a **thermal conduit** for the heat exchange system, as well as integrates the **cooling system** that may be put into the mold halves.
- The **mold halves** are mated and aligned, and the part surfaces are finished for surface quality.
- Using extended cycle times and a release agent, **numerous parts** can be made by directly injecting the final thermoplastic into the ACES mold core and cavity halves using a standard injection-molding machine.

cooling system



mold halves

Direct AIM tooling

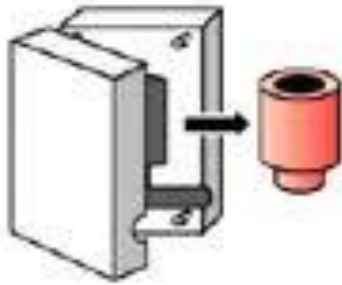


Ejector pins:
For ejecting
finished part

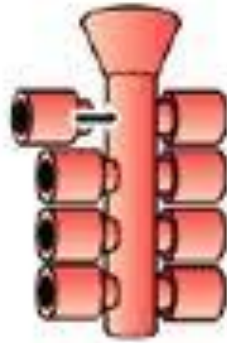
Direct AIM tooling

Side note, not to be studied for this course

The Basic Steps in the Investment Casting Process



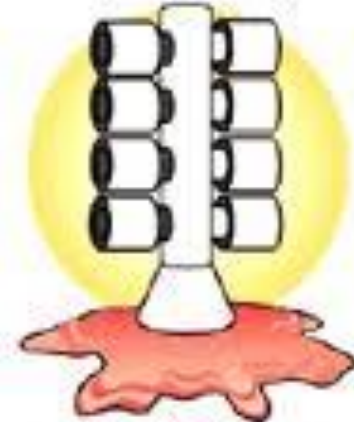
Wax Injection



Assembly



Shell Building



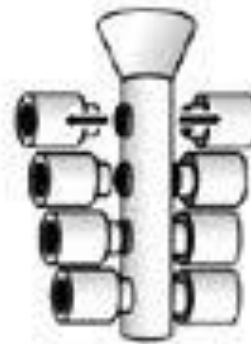
Dewax/Burnout



Gravity Pouring



Knock Out



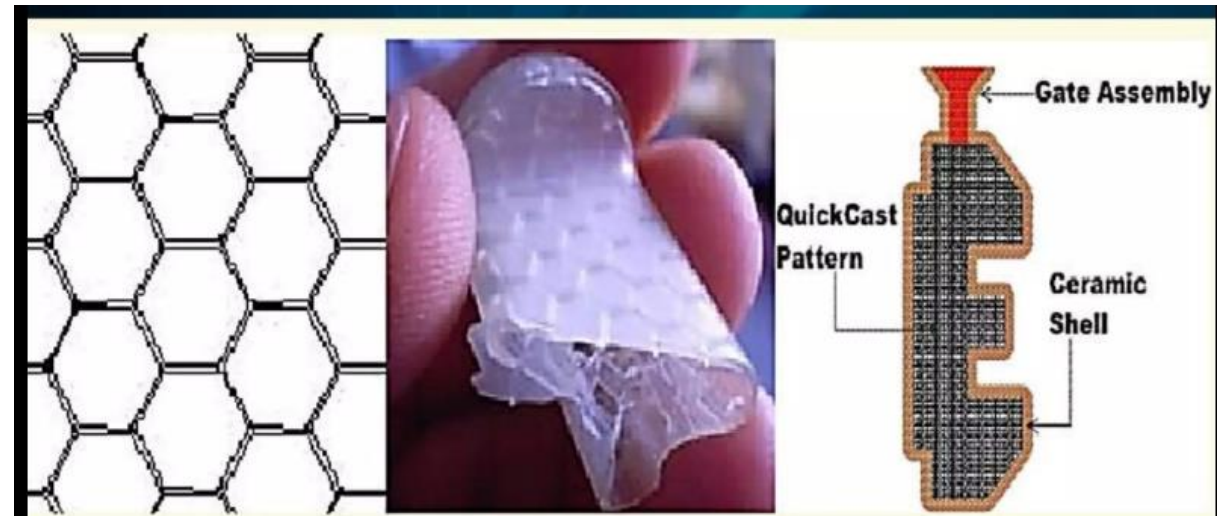
Cut-off

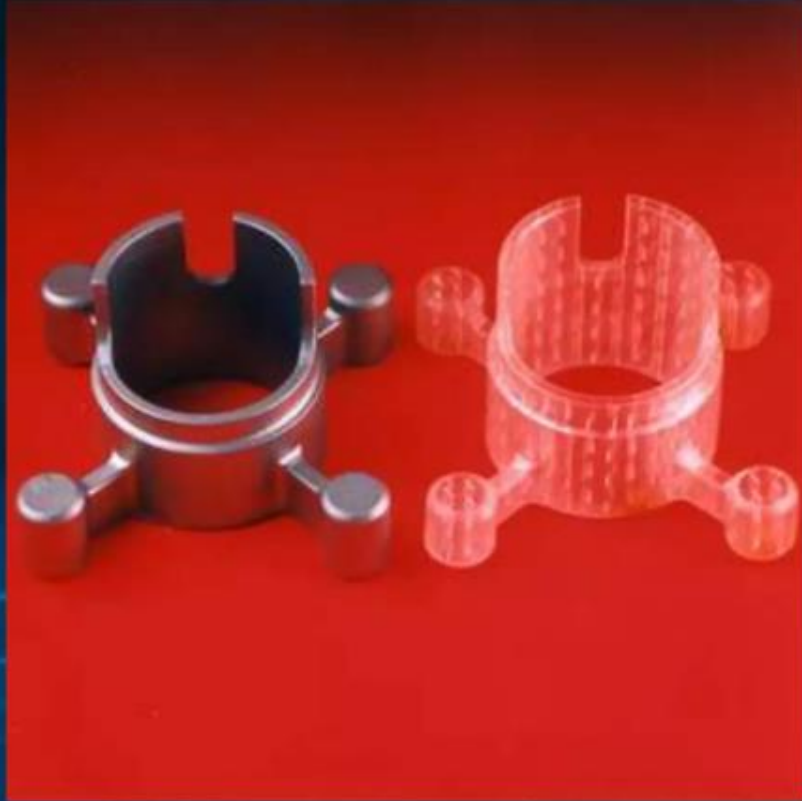


Finished Castings

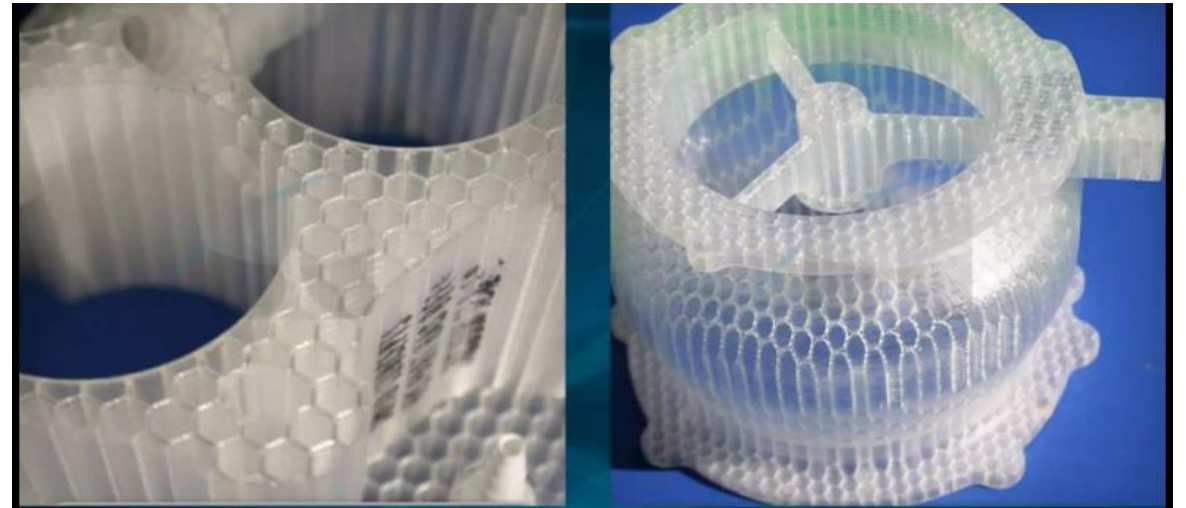
QuickCast

- A **Stereolithography QuickCast pattern** is created from an STL file.
- The pattern is leak tested to make sure it is air tight.
- An investment caster is chosen (based on experience & material required).
- QuickCast pattern is given to the caster.
- Caster puts part through ceramic coating process and performs firing procedure to burn out SLA pattern.
- Metal is poured into the fired ceramic shell.
- Ceramic shell is broken off to reveal metal part.





QuickCast Stereolithography resin model and steel casting



Investment casting patterns made by QuickCast

Copper polyamide Tooling

- Copper Polyamide material is a heat resistant, thermally conductive **composite of copper and plastic**
- Copper polyamide is a new metal plastic composite designed for short run tooling applications (**100 to 400 parts**) from common plastics.
- Tooling inserts are produced directly in the **SLS** machine with a **layer thickness of 75 μm** .

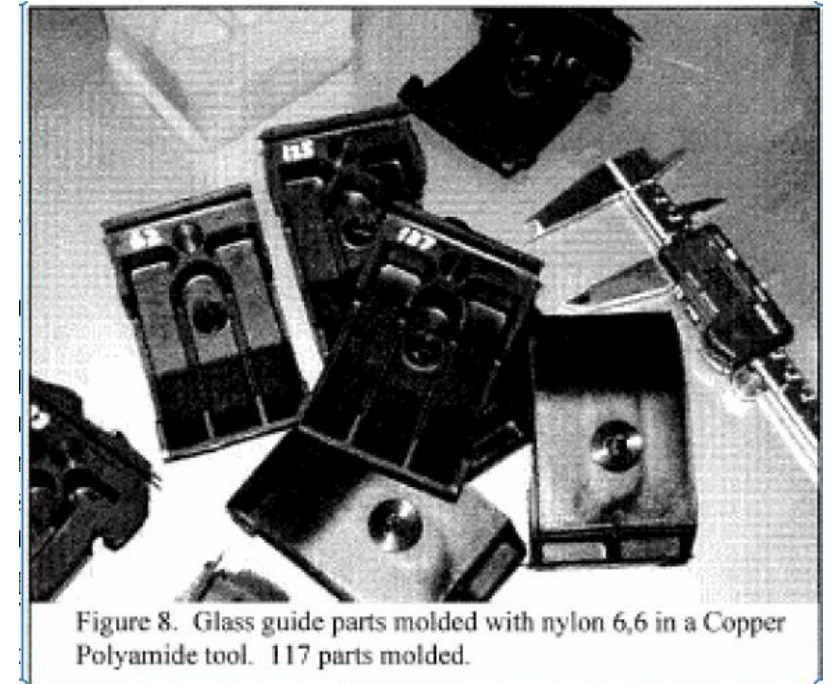


Figure 8. Glass guide parts molded with nylon 6,6 in a Copper Polyamide tool. 117 parts molded.

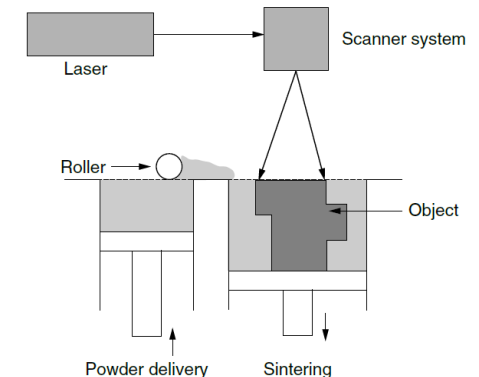
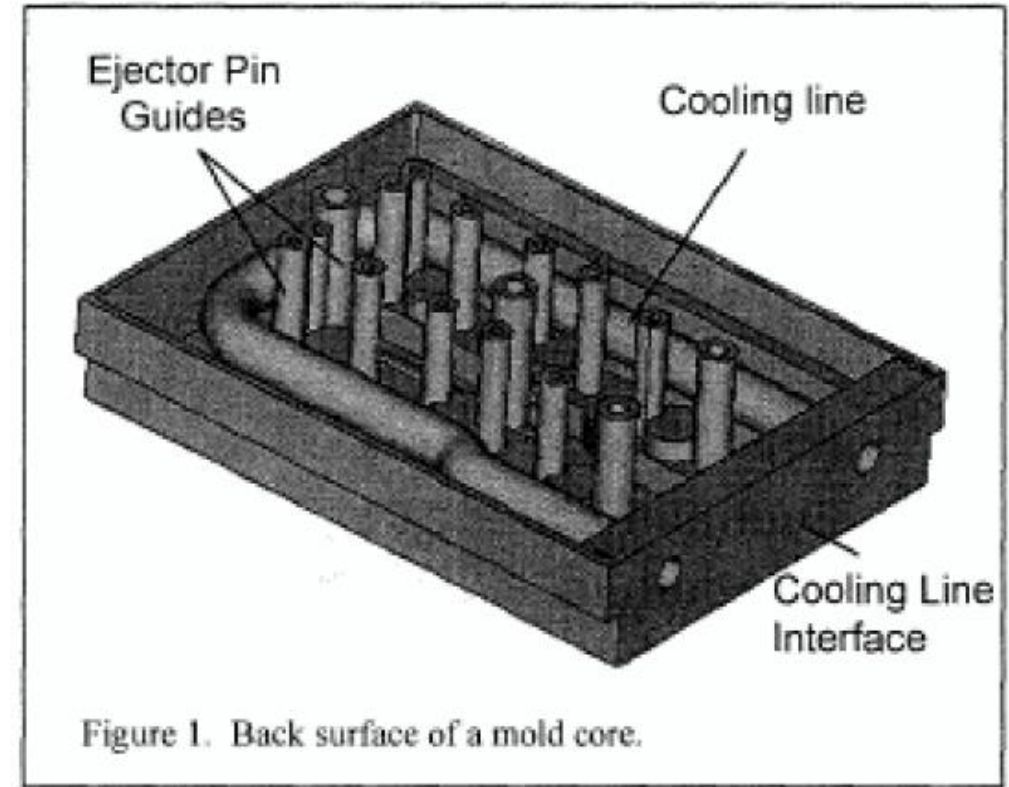


Figure 8.10 Selective laser sintering.

- **Subsequent finishing** is necessary before their integration in the tool base.
- No furnace cycle is required and unfinished tool insert can be produced in a day.
- During the CAD stage, Copper polyamide **inserts** are shelled.
- **Cooling lines, ejector pin guides, gates and runners** are included in the design and built directly during the SLS process.
- Then the insert surface are sealed with epoxy and finished with sand paper and finally the shell inserts are packed up with a metal alloy.



- Inserts produced from copper polyamide are easy to machine and finish.
- Heat resistant and thermal conductivity are better in most plastic tooling materials.
- The cycle times of moulds employing copper polyamide inserts are similar to those of metal tooling.

Sand Casting

- Rapid prototyped models can also be used as patterns for sandcasting.
- A sand pattern consists of sand mixed with bonding agents, which is all contained within a flask.
- In sand casting, a model is used to make an impression in two halves, or the cope and drag of the sand pattern.
- Molten metal is cast into this sand impression, allowed to cool, then pulled from the cavity and the process is repeated.
- An RP pattern can sometimes withstand several applications of this technique before becoming distorted or damaged, therefore several castings can be made from the rapid prototyped model.

- Although a slight loss in dimensional accuracy may occur as compared to investment casting, this process can be used to create several test articles for preliminary tests. Also, with allowances for final machining in the model, it can produce directly usable hardware in small to large quantities.
- The major drawback to sand casting is the limitations on part complexity. If parts have sharp edges, internal structures or protrusions with inappropriate draft, often times that geometry cannot be sand cast. The parts must be shaped so that the pattern can be pulled from the sand mold without disrupting the surface.